Damp Heat Induced Degradation of Silicon Heterojunction Solar Cells With Cu-Plated Contacts

Joseph Karas, Archana Sinha, Viswa Sai Pavan Buddha, Fang Li, Farhad Moghadam, Govindasamy TamizhMani, Stuart Bowden, and André Augusto

***Abstract*—Damp heat exposure is one of the most stringent environments for testing the durability of solar cells in packaged modules. Damp heat stresses and induces a variety of degradation modes in solar cells and modules: for example, moisture-induced corrosion of electrodes and interconnections, deterioration of polymeric materials, and/or thermally activated diffusion processes. To screen for these and other potential degradation modes, we subject one-cell modules containing silicon heterojunction (SHJ) solar cells with Cu-plated contacts to extended damp heat tests at 85 °C/85% relative humidity. SHJ cells were laminated with two common encapsulants: ethylene vinyl acetate (EVA) and polyolefin elastomer (POE), and two constructions: glass–backsheet and glass–glass. We observe degradation in all components of solar cell maximum power (P_{MP}): current, voltage, and fill factor, and find evidence of increased carrier recombination and nonideal diode behavior with increasing stress. For glass–backsheet constructions, EVA samples generally degrade more than POE by a factor of approximately 1.5x P_{MP}, and the different encapsulants produce different degradation patterns. Similar trends are observed in glass–glass modules, but to a lesser degree. In a different experiment, we observe a decrease in effective minority carrier lifetime of nonmetallized SHJ precursors measured after damp heat. This implies that some degradation unrelated to the contacts is to be expected and confirms the observation of increasing recombination.**

***Index Terms*—Copper (Cu) plated contacts, damp heat (DH), encapsulant, fill factor (FF), reliability, silicon heterojunction (SHJ).**

I. Introduction

SILICON heterojunction (SHJ) solar cells outperform standard commercial crystalline silicon (c-Si) solar cells, such as aluminum-back-surface-field and passivated emitter and rear cells, by operating at higher voltages [1]. Higher voltages are a direct result of the amorphous/crystalline silicon (a-Si/c-Si) heterostructure, which provides superb surface passivation and physically separates the absorber from highly recombination-active metal contact areas leading to rates of surface recombination two orders of magnitude lower than standard c-Si cells [2], [3]. These high efficiencies, an inherently bifacial structure, and an adaptable and relatively simple manufacturing process make them an excellent platform technology for future photovoltaic (PV) concepts [4].

One challenge for SHJ lies in the metallization. As the high performance of SHJ devices relies on thermally sensitive a-Si passivation layers, the metallization process (which for typical c-Si cells involves high-temperature firing over 700 °C) must take place below ∼250 °C. This temperature restriction makes it impossible to form low line resistivity contacts using traditional silver (Ag) screenprinted pastes. Low curing-temperature Ag pastes are available, but the resistivity is higher and requires greater Ag content [5]. Additionally, cell interconnection with low-temperature Ag paste is more challenging and alternative interconnection (e.g., conductive adhesives or multiwire) are often used instead of traditional soldering. In recent years, copper (Cu) metal plating has emerged as a metallization alternative that may solve these issues. Cu contacts can be plated at low temperature, and have a bulk resistivity equivalent to bulk Cu and several times lower than SHJ-compatible screenprinted Ag pastes. A number of high-efficiency and record results have been recorded on SHJ cells with Cu-plated contacts [6]–[8].

This has motivated interest in Cu-plated contacts for c-Si solar cells for some time; however, they have not been widely adopted. A number of barriers to plated contacts have been identified for diffused junction cells, including concerns about long-term degradation and reliability in the field [9]. Some of these barriers also apply when Cu-plated contacts are applied to SHJ cells.

For SHJ cells to be economically viable versus other technologies, their high performance, which relies on high-quality surface passivation, must be maintained over the service life of the system. Any new reliability or durability risks must be understood and mitigated. Presently, there is a limited existing literature on both accelerated degradation studies and field performance of SHJ cells in general. Several longer-term field studies have been performed, noting that open-circuit voltage (V_{OC}) can degrade for SHJ modules [10]. This possibly indicates

Manuscript received June 3, 2019; revised August 8, 2019; accepted September 3, 2019. Date of publication October 21, 2019; date of current version December 23, 2019. This work was supported in part by the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (DOE EERE) under Solar Energy Technologies Office (SETO) Agreement 32509 for the Durable Module Materials Consortium (DuraMAT), and in part by the Engineering Research Center Program of the National Science Foundation (NSF) and the DOE EERE under NSF Cooperative Agreement EEC-1041895 (QESST). *(Corresponding author: Joseph Karas.)*

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Digital Object Identifier 10.1109/JPHOTOV.2019.2941693

loss of surface passivation or increased bulk recombination, and differs from what is typically seen in other c-Si modules [10], [11]. At present, the physical mechanisms of these losses remain conjecture.

At this date, few literature studies on accelerated reliability testing of SHJ cells or modules exist, and even fewer explicitly consider SHJ cells with Cu-plated contacts. Adachi *et al.* [12] studied SHJ cells with Cu-plated contacts in damp heat (DH) tests and achieved promising results, in which single-cell mini-modules underwent thousands of hours of DH exposure with minimal degradation. However, this article was primarily designed to test the role of a SiO_X capping layer in preventing moisture or Na^+ ion degradation rather than the Cu-plated contact specifically [12].

Cu-plated contacts in general are thought to introduce several new forms of reliability risks to traditional c-Si PV cells [9]; and some of these concerns are also relevant for Cu-plated contacts applied to SHJ cells. Additionally, SHJ cells are being considered for use in novel and nonstandard module constructions and deployment situations, including bifacial modules with glass–glass (G–G) construction or transparent backsheets. SHJ cells have a lower temperature coefficient for power than other c-Si solar cells, so may be deployed more frequently in hot climates. Altogether, these features may mitigate some traditional reliability risks, but may also introduce new risks [13].

In this article, we have fabricated SHJ modules with Cu-plated contacts and performed accelerated DH tests (85 °C and 85% relative humidity) to the IEC 61215 protocols, and we extend the test duration beyond the standard 1000-h duration. The objective was to examine the temperature and moisture-induced degradation mechanisms that may be introduced with Cu-plated contacts on SHJ cells. We also examine SHJ precursor cells without metallization to understand how SHJ passivation and lifetime may be affected by DH stress.

II. Experiment

This article was carried out on two sets of SHJ cells with Cu-plated contacts. The first set was sourced from a pilot-scale laboratory process [14] (baseline efficiency ∼20%), while the other set was sourced from a premium commercial supplier (efficiency ∼23%) to compare with our research-cell samples. The lab-processed cells used a sputtered Ag seed layer, with grid patterning performed via dry film photoresist. All cells used were bifacial (even those eventually manufactured into monofacial modules), capable of absorbing light from both front and rear sides. Direct Cu electroplating was performed simultaneously on both sides of the cell with target finger height of 12 μm. Fingers were then capped with a ∼200 nm thick solderable tin (Sn) layer. Solar cells were tabbed manually using rosin flux and PV ribbon, then laminated between 3 mm thick, 180 mm^2 soda-lime solar glass (Solite), and encapsulants- either ethyl vinyl acetate (EVA) or polyolefin elastomer (POE). On the rear, we used either a breathable, Tedlar-based PV backsheet, or a second piece of 3 mm glass. Lamination conditions were according to each encapsulant manufacturer's recommendations.

Samples were subject to DH in environmental test chambers following IEC guidelines for 1800–3000 cumulative hours. Samples were removed for characterization every 500–1000 h. Characterizations included illuminated current–voltage (*I–V*), Suns-V_{OC} (Sinton Instruments FCT-450), electroluminescence (EL, Si CCD camera with 850 nm highpass filter and 5 A current injection), and photoluminescence (PL, Si CCD camera with 850 nm highpass filter and ∼0.2 suns excitation at 808 nm), among others. Suns-V_{OC} measurements, which plot the V_{OC} of the device against current-calibrated illumination, are a useful way of understanding different contributions to cell degradation over time. The pseudo-parameters derived from the Suns-V_{OC} curve provide a true measure of series resistance at maximum power as well as recombination-related parameters that reduce fill factor (FF), which are difficult to discern from *I–V* curves alone [15], [16].

III. Results and Discussion

A. *Effect of Glass–Backsheet (G–BS) Versus G–G Module Construction*

Previous studies of cell-level degradation in DH have identified corrosion of front metallization as a characteristic degradation mode [17]. It is thought to be caused by water permeation through the backsheet and around the front of the cell. The front encapsulant is hydrolyzed, which creates acidic products that corrode front metal, first at the perimeter of the cell, then advancing toward the center with continued DH exposure. This mode of degradation has been confirmed in various ways, although it has primarily been studied for traditional (non-SHJ) c-Si cell structures [17]. As the Cu-plated contacts on the front and rear of these bifacial cells are the same, any acid-induced degradation of the contacts may occur sooner, on the rear side of the module.

To simultaneously test a module with more limited moisture intrusion, we simply use glass in lieu of a rear backsheet. This is of course also relevant to SHJ cells because of their intrinsic bifaciality. In G–G module constructions, moisture ingress is limited to cell edges, which must travel further laterally through the encapsulant to reach the cell and greatly increases the time required to observe moisture-induced degradation [18]. In previous DH studies that include SHJ modules with G–G construction or some other rear moisture barrier, the traditional moisture-dependent cell edge degradation mode does not occur [19], [20].

Fig. 1 shows *I–V* degradation of SHJ cells, two encapsulated with a permeable backsheet (G–BS), two with rear glass (G–G). The encapsulant in both cases was EVA, and no edge seals or other moisture barriers were employed. Degradation in maximum power (P_{MP}) trends very similarly for both sets of samples, with only slightly greater degradation in G-BS samples. After 1000 h, G-BS lost 2.3% P_{MP} compared to 1.8% for G–G samples. After 2500 h, G–BS and G–G samples were at 6.4% and 4.8% P_{MP} loss, respectively. The differences in *I–V* degradation are slight between the two modules, with all modules undergoing similar loss in short-circuit current (I_{SC}), open-circuit voltage (V_{OC}), and FF. I_{SC} and FF loss begin earlier, after perhaps 1000 h, with substantial V_{OC} loss occurring after 2500 h, and perhaps a bit earlier for G–BS samples. FF loss is responsible for 50%–80% of the overall degradation in each sample, with I_{SC} and V_{OC} loss each responsible for 10%–30%. Relative increases

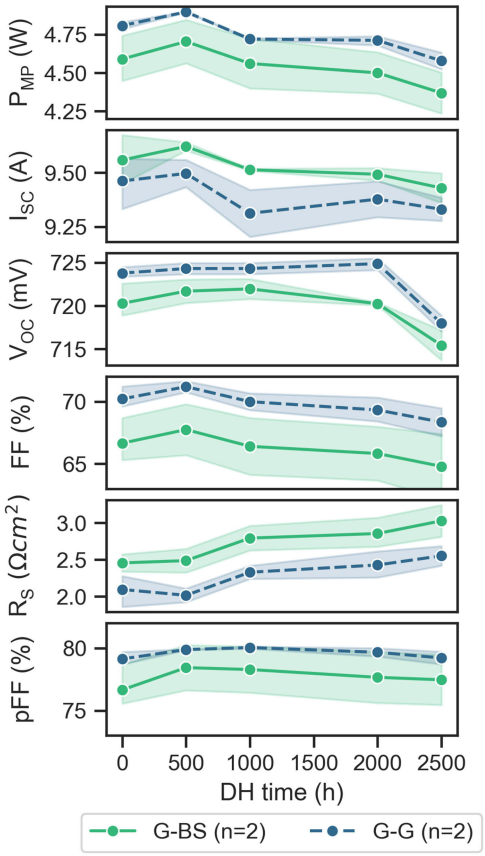

Fig. 1. *I–V* measurements of Cu-plated SHJ modules with G–BS and G–G construction with increasing DH test time. Number of samples of each type are indicated by "*n*," solid markers are the mean of the sample set and shaded regions indicate the 75% confidence interval.

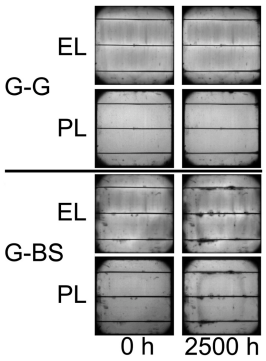

Fig. 2. EL and PL images of selected G–G and G–BS samples before at 0 and 2500 h DH exposure.

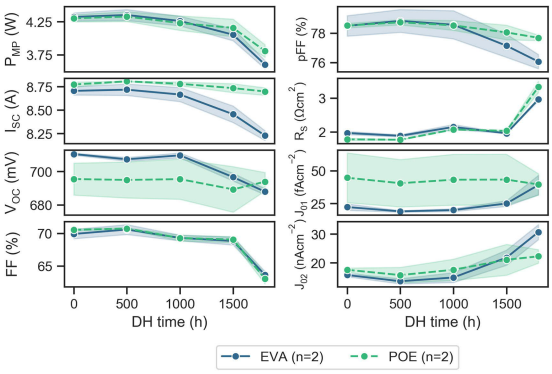

Fig. 3. *I–V* measurements (left) and selected Suns-V_{OC} parameters (right) of Cu-plated SHJ modules with increasing DH exposure, encapsulated with EVA and POE. All samples have a permeable rear backsheet. Number of samples of each type are indicated by "*n*," solid markers are the mean of the sample set and shaded regions indicate the 75% confidence interval.

in series resistance (R_S) are nearly identical for both sample types, as are reductions in pseudo FF (pFF). The pFF parameter is the FF of the Suns-V_{OC} curve (which is unaffected by series resistance), rather than the *I–V* curve (which includes series resistance), and is a general indicator of diode quality. Comparison of pFF and FF is, therefore, a way to evaluate R_S-related losses [21]. Overall, loss of pFF describes approximately 10%–40% of the overall FF loss, with R_S responsible for the remainder. The most obvious difference between the two types of samples is shown in EL images in Fig. 2. G–G samples are largely unchanged, with only a small amount of darkening at the edges after 2500 h DH. G–BS samples, in contrast, have new dark areas appearing along busbars, and a dark ring-like pattern at the center. The same dark area is also visible in PL images, and its presence in both PL and EL suggests it is related to increased recombination, not simply increased R_S. At present, the physical origin of each of these dark regions is unclear, as is the precise manner and extent to which they affect cell performance.

B. *Effect of Encapsulant—EVA Versus POE*

In prior studies, the corrosion of c-Si cell contacts in DH can be averted by choosing an encapsulant material that does not evolve acidic products with moisture exposure, for example, POE [22]. We fabricated G–BS samples from SHJ cells, two with EVA and two with POE. The *I–V* and Suns-V_{OC} results after 1800 h of DH exposure are shown in Fig. 3. After 1000 h, the degradation is similar and slight, with only ∼1% P_{MP} loss. However, after 1800 h, POE samples lose 11% P_{MP} while EVA samples lose 16% P_{MP}, on average. For EVA samples, all of I_{SC}, V_{OC}, and FF decline. Most of the decline in I_{SC} is an extrinsic factor caused by encapsulant discoloration, which correlates with changes in yellowness index and ultraviolet fluorescence measurements (not shown here). Slight V_{OC} loss for EVA samples begins at ∼1500 h (confirming the results in Fig. 1), and substantial FF loss for both EVA and POE samples occurs at 1800 h.

pFF loss is slightly greater in EVA samples, and explains roughly one-third of their overall FF loss. pFF loss explains roughly one-tenth of overall FF loss in POE samples. The rest is explained by increased R_S, which is similar for both sample types. Otherwise, J_{01} and J_{02}, the parameters obtained by fitting the two-diode model to the Suns-V_{OC} curve with fixed ideality factors $n = 1$ and $n = 2$, respectively, increase somewhat more for EVA samples than for POE samples. The changes in diode quality parameters (pFF, J_{01}, J_{02}) begin at 1500 h, corresponding with V_{OC} loss. All of these parameters suggest

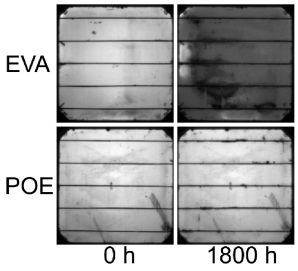

Fig. 4. PL images before and after 1800 h DH exposure for two selected samples from the data shown in Fig. 3.

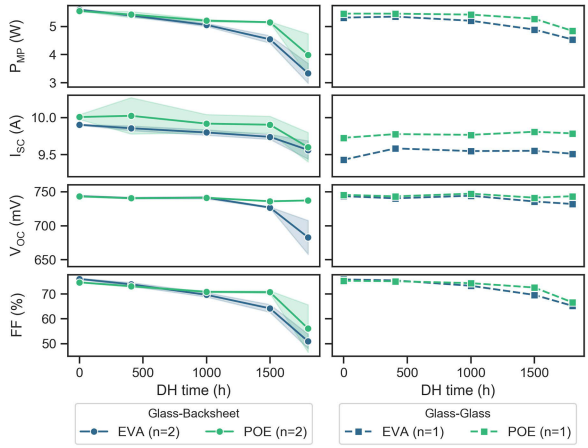

Fig. 5. *I–V* measurements with increasing DH time, of modules with commercial Cu-plated SHJ cells with G–BS (left) and G–G construction. Number of samples of each type are indicated by "*n*," solid markers are the mean of the sample set and shaded regions indicate the 75% confidence interval.

increased recombination affecting the shape of the *I–V* curve near the maximum power point. J_{01} is classically interpreted as recombination contributions from the bulk and emitter of the cell, while J_{02} is classically interpreted as depletion region recombination. However, in this case (especially as this is a heterojunction device), the increase in J_{02} is more likely due to localized recombination in degraded regions of the cell [23]. PL images shown in Fig. 4 show POE samples with just a few dark areas located along busbars, while EVA samples darken generally, indicating reduced carrier lifetime, and again have a large dark ring-like area in the cell interior, similar to Fig. 2. POE seems to substantially mitigate degradation of this type.

C. SHJ Cell Degradation in DH—Commercial-Quality Samples

In order to control for the batch-to-batch performance variability inherent in the lab-produced cells in Sections III-A and III-B, we repeated the experiments with Cu-plated SHJ cells sourced from a commercial supplier [24]. *I–V* and Suns-V_{OC} degradation of these samples are presented in Figs. 5 and 6. In general, these higher efficiency samples seem more vulnerable to degradation but exhibit similar trends to the data shown in Sections III-A and III-B. G–BS samples encapsulated with EVA degrade the most, with 9% P_{MP} loss after 1000 h, and 40% loss

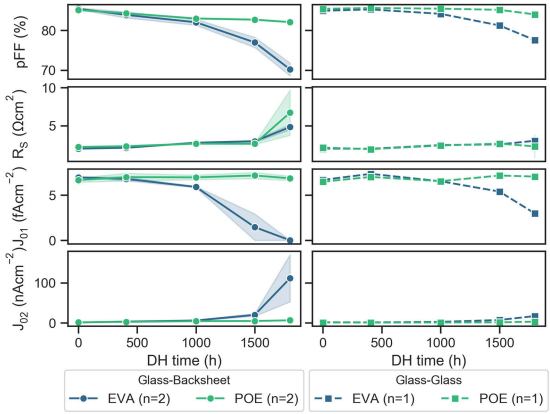

Fig. 6. Suns-V_{OC} parameters with increasing DH time of the modules in Fig. 5. Number of samples of each type are indicated by "*n*," solid markers are the mean of the sample set and shaded regions indicate the 75% confidence interval.

after 1800 h. G–BS with POE lose 6% and 35% P_{MP} after 1000 and 1800 h, respectively, however, at 1800 h the P_{MP} loss seems to be driven by large increases in R_S for one of the specimens, likely attributable to extrinsic module interconnection failure. In both encapsulants, losses are suppressed when G–G modules are employed. Again, all of I_{SC}, V_{OC}, and FF decline over time, with substantial pFF declines after ∼1500 h DH for EVA specimens. For EVA specimens, most of the overall FF decline can be attributed to this pFF loss; this is not true for POE. For EVA samples, the pFF declines coincide with increases in J_{02}, while J_{01} actually decreases. As in Section III-B, these changes should not be interpreted to have classical meaning, but as fitting parameters that indicate the increasing and spatially nonuniform nonideality of the Suns-V_{OC} curve. R_S increases substantially for G–BS specimens of both encapsulant types at 1800 h. Again, we conclude that at the cell level, the overall power loss is a combination of increasing R_S and increasing recombination and/or non-ideal diode behavior near the maximum power point. The losses from recombination and nonideality are greater in EVA-encapsulated samples.

D. Lifetime Degradation in DH

To study degradation unrelated to the metal contact, we also examined unmetallized cells, that is, precursor SHJ cells lacking any metal contact. We passivated n-type Cz c-Si wafers with a-Si and then deposited indium tin oxide transparent conducting layers on both sides. Samples were encapsulated in EVA with a Tedlar backsheet and subjected to DH. Lifetime measurements were performed on the encapsulated samples via the quasi-steady-state photoconductance method (QSSPC, Sinton WCT-120). Note that to obtain accurate measurements in this way, new calibration constants must be generated to account for the reduced QSSPC signal with intermediate backsheet and encapsulant layers between the sample and the sensor [25].

Effective lifetime (τ_{eff}) of five wafers with increasing DH exposure is shown in Fig. 7. In general, lifetime decreases with continued DH exposure, with more substantial degradation occurring after 1500–2000 h DH. Lifetime curves of wafer B are

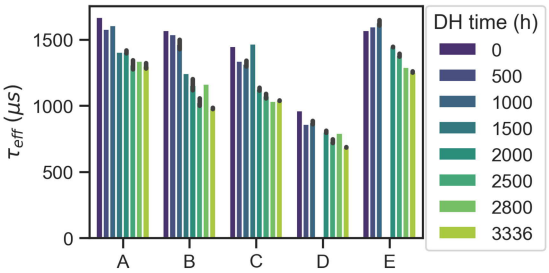

Fig. 7. Effective lifetime at minority carrier density $= 10^{15}$ cm^{-3} of five unmetallized SHJ precursor wafers with increasing DH duration. Samples were encapsulated in EVA with a permeable backsheet. Black error bars indicate the standard deviation of multiple measurements.

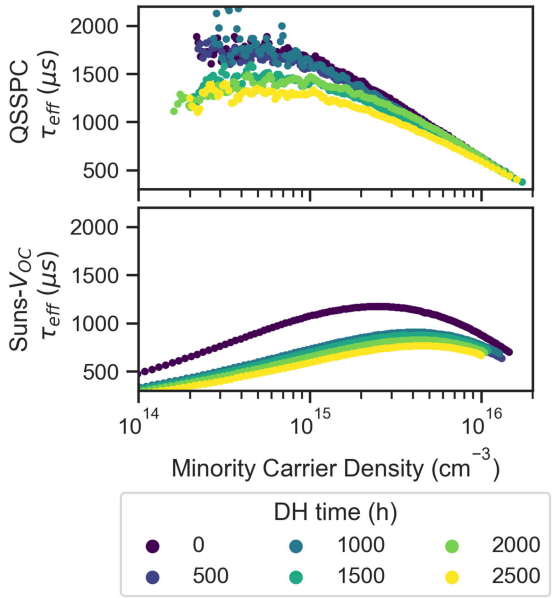

Fig. 8. Effective lifetime curves versus minority carrier density for (top): an unmetallized SHJ precursor wafer (wafer B in Fig. 7); and (bottom): a Cu-plated SHJ cell (the G–BS sample from Fig. 2). Color indicates DH test time. Unmetallized wafer lifetime is measured via QSSPC, while cell lifetime is measured via Suns-V_{OC}.

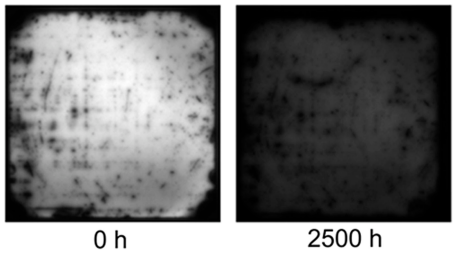

Fig. 9. Photoluminescence images before and after 2500 h DH, for wafer B in Fig. 7, which is also the sample for which lifetime curves are shown in the top panel of Fig. 8.

shown in Fig. 8, as well as lifetime computed from Suns-V_{OC} for an EVA-encapsulated G–BS cell (the G–BS sample of Fig. 2). For both samples, lifetime trends downward with longer DH exposure, providing evidence that in EVA, cell lifetime degrades irrespective of metallization or other cell components. PL before and after exposure (see Fig. 9) show increased edge darkness, increased occurrence of dark points in the cell interior, and a

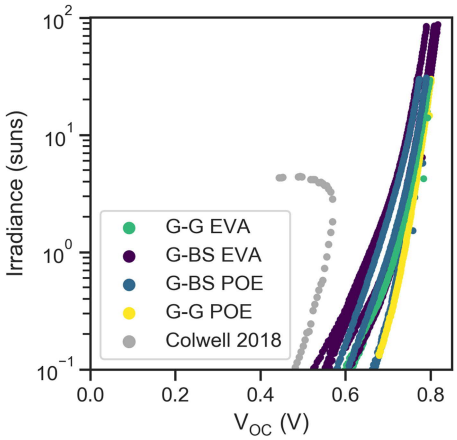

Fig. 10. V_{OC} versus irradiance, including irradiance >10 suns, for all SHJ cells in this article at the end of their DH exposure. Color indicates module construction (G–G or G–BS) and encapsulant type (EVA or POE). An example of the voltage bend-back effect from [27] is shown in gray.

general reduction in luminescence indicating reduced lifetime. Dark damage spots present initially are artefacts of wafer sawing, texturing, and handling, and not related to DH degradation [26].

E. Suns-V_{OC} at High Irradiance

To examine one final mechanism known to affect Cu-plated contacts, we performed Suns-V_{OC} measurements at very high irradiance (>10 suns). Colwell *et al.* [27] previously showed that c-Si solar cells with Cu-plated contacts can undergo Schottky barrier formations within the contact metal [27]. This happens when Cu in the contact migrates into capping layers with thermal stress, creating voids in the contact stack. The Schottky barriers can reduce FF, pFF, and potentially V_{OC}, and make it difficult to interpret Suns-V_{OC} data accurately. Schottky contact behavior is readily identifiable by "bend-back" in V_{OC} at irradiance >3 suns. Fig. 10 shows V_{OC} versus light intensity for all the modules in this article at the end of their final DH exposure (1800–2500 h). Data from [27] is also shown, although because of differences between this article in the cell architecture (BSF versus SHJ), capping layer (plated Ag versus Sn), and the use of nonencapsulated cells, this data are shown purely as an example of voltage bend-back. As we do not observe the same Schottky behavior, we are confident this degradation mechanism does not play a significant role in the cell degradation we observe.

IV. Conclusion

In this article, we observed that Cu-plated SHJ solar cells in packaged modules do not degrade in the same ways as traditional c-Si cells when subject to DH. We observed loss in V_{OC}, I_{SC}, and FF. Increase in R_S explains only part of the FF loss, while Suns-V_{OC} parameters provide evidence of increasing recombination and increasing nonideal behavior with longer DH exposure. Increases in Rs explain may be attributable to degradation of the Cu-plated contacts, some degradation unrelated to the contact was also observed. This can be further studied by encapsulating and stressing unmetallized lifetime

samples. Furthermore, the encapsulating material plays a role in determining the magnitude, if not also the primary mechanism, of degradation, with POE generally resulting in less degradation than EVA. Limiting moisture ingress with G–G construction suppresses but does not eliminate lifetime degradation. In EVA with a permeable backsheet, lifetime loss may result in observable P_{MP} and V_{OC} degradation within the standard (1000 h) DH test duration. The observed degradation could not be attributed to Schottky barrier formations within the Cu-plated contact.

Both SHJ cells and the specific case of SHJ cells with Cu-plated contacts potentially introduce new PV reliability risks and, therefore, warrant further study to understand the severity and impacts of these risks.

Acknowledgment

The authors would like to thank S. Herasimenka (Regher Solar, LLC) for insightful discussion and A. Srinivasa and P. Balaji (Arizona State University) for providing samples for this work.

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